[+] HEALTHCARE INNOVATION • AI OPERATING SYSTEM

Zero Wait Times.
Universal Patient Inclusion.

An AI-powered smart hospital operating system integrating on-device prescription OCR, 6-language voice AI, visual symptom triage, and live queue proximity alerts.

INNOVATION TITLE
ArogyaNext AI

INSTITUTION
Mohamed Sathak Engineering College

LIVE DEPLOYED SYSTEM
https://arogyanext.in

Critical Gaps in Outpatient Healthcare

Over 70% of Indian OPD visits suffer from prolonged delays, demographic exclusion, and disjointed care.

2-4 Hrs

Waiting Room Congestion

Patients endure exhausting, unmonitored queues with zero real-time progress visibility, creating massive lobby crowding and infection risks.

65%+

Digital & Language Divide

Elderly, rural, and illiterate patients cannot navigate complex text-heavy apps, causing severe triage mistakes and wrong-specialist visits.

80%

Lost Paper Prescriptions

Physical slips get lost, damaged, or are illegible, forcing doctors to re-diagnose blind without access to past medication history.

40 Mins

Pharmacy Queue Friction

Post-consultation medicine collection causes secondary bottlenecks and pharmacy revenue leakage away from the hospital.

The Complete Smart Clinic Ecosystem

ArogyaNext connects the entire loop from the patient's home to the doctor's cabin and in-house pharmacy.

Accessible Patient Mobile App

On-device ML Kit OCR scanner for old paper prescriptions, multi-lingual voice booking (6 Indian languages), visual body triage, and live queue proximity voice alerts.

Android Kotlin | ML Kit OCR | Regional Speech

Doctor Workspace & Smart Pad

Instant access to past scanned OCR prescriptions, 1-click patient calling, automated lobby TV voice triggers, and calibrated letterhead e-prescribing.

Past Rx Auto-Fill | Digital Ink Sign | Letterhead Calibrator

TV Signage & E-Pharmacy Hub

Smart TV lobby signage announces live tokens aloud via text-to-speech. Digital prescriptions beam instantly to the in-house pharmacy for zero-wait checkout.

Web TTS Speech | Neon TV Queue | Instant Pharmacy Sync

Proprietary Innovations & IP Strengths

Engineered specifically for demographic inclusion, privacy, and high-speed clinical execution.

On-Device OCR Scanner

Powered by Google ML Kit. Patients photograph paper slips; edge AI parses previous medications without uploading private images to external clouds.

6-Language Voice AI

Native voice recognition in English, Hindi, Tamil, Telugu, Marathi, and Bengali for total regional accessibility without typing barriers.

Visual & Audio Triage

Interactive 4-region body map, intuitive symptom cards, color-coded pain scale, and audio voice prompt guidance for senior and non-literate patients.

Proximity Voice Alerts

Calculates live wait times (~5 min/token). When 0–4 spots away, speaks aloud to patient with interactive check-in (`[ Proceed ]` / `[ Cancel ]`).

Synchronized Multi-Tenant Cloud Architecture

Cloud Firestore real-time snapshot listeners ensure sub-50ms synchronization across all facility touchpoints.

01

Patient Check-In

Patient books via Android app (voice/OCR/body map) or front-desk receptionist. Live digital token (e.g. S-24) is issued with estimated arrival time.

02

Queue & TV Sync

Waiting Room TV updates in real-time. Patient receives countdown proximity alerts on their smartphone when approaching consultation.

03

Doctor Consultation

Doctor calls patient; TV announces token aloud. Doctor reviews ML Kit scanned past Rx, writes e-prescription, and e-signs on tablet pad.

04

E-Pharmacy Packing

Prescription syncs to hospital pharmacy immediately. Medicine is bagged and ready before patient walks to the dispensary counter.

Massive Addressable Healthcare Market

Targeting India's rapidly modernizing outpatient clinic and mid-size private hospital network.

70,000+

Private Hospitals & Clinics

Primary market across Tier-1, Tier-2, and Tier-3 cities in India seeking affordable digital queue management without complex legacy ERP costs.

1.2M+

Registered Practicing Doctors

Clinicians overburdened by administrative overhead and paperwork who benefit from automated letterhead prescription generation and OCR past notes.

1.4B

Citizen Demographic Reach

Inclusive visual & voice design empowers the bottom of the demographic pyramid: senior citizens, non-English speakers, and rural migrant patients.

Scalable B2B SaaS Monetization

Low barrier to entry with recurring subscription pricing and zero expensive hardware requirements.

Base Clinic License

₹999/month

Full access to front desk receptionist scheduler, waiting room TV lobby signage, e-pharmacy routing, and administrative analytics hub.

- Includes 1 Doctor
- Unlimited Patients

Per Doctor Seat Add-On

+₹500/doctor/mo

Doctor workspace portal, Smart Tablet prescription pad, on-device OCR paper slip auto-parsing, and letterhead calibration engine.

- Doctor Portal
- Tablet Pad

Pharmacy Value Addition

Revenue Retention

Eliminates 30% prescription leakage to outside pharmacies by auto-routing medicines internally, paying for the software within the first 3 days of every month.

- Zero Leakage
- High ROI

Execution Roadmap & Institutional Backing

Ready for field pilots and clinical deployment.

Development Team

Student Innovators: Mohamed Sathak Engineering College
Domain: AI/ML, Cloud Systems & Mobile Health Architecture

Live Production Deployments:

- `Web Portal: https://arogyanext.in`
- `Android App: Kotlin / Google ML Kit`

Growth & Deployment Roadmap

- **Phase 1 (Current):** Complete working prototype across Web & Android.
- **Phase 2 (Q2 2026):** Pilot test in 5 local outpatient clinics in Tamil Nadu.
- **Phase 3 (Q3 2026):** Integrate with Ayushman Bharat Digital Mission (ABDM).
- **Phase 4 (Q4 2026):** Expand regional language voice models to 12 Indian dialects.